

Hemil Patel

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SUMMARY

4 years of driving growth through data-driven insights and process improvements, blending business acumen with technical expertise. Delivered results in complex business environments, fostering efficiency, cost reduction, and productivity gains. Skilled in stakeholder engagement, team collaboration, and effective communication, driving informed decision-making and sustainable success.

EDUCATION

CALIFORNIA STATE UNIVERSITY <i>Master of Science in Computer Science</i>	SACRAMENTO, California <i>January 2021 – December 2023</i>
UKA TARSADIA UNIVERSITY <i>Bachelor of Technology in Computer Engineering</i>	India <i>May 2016 – May 2020</i>

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, SQL, HTML, CSS
- **Frameworks:** Hadoop Framework, Pandas, Keras, Sci-Kit Learn, Django, Jupyter Notebook.
- **Tools:** Pre-silicon validation, SOC architecture, Power Controller IP, Bootloader IP, FPGA validation, RTL simulation, UEFI, BIOS, Computer architecture, Git, JTAG, Oscilloscopes, Logic Analyzers.
- **Business Analysis:** Requirements Elicitation, Business Process Modeling, Stakeholder Engagement, Agile Methodologies, Business Requirements Document (BRD), Functional Requirements Document (FRD), Software Requirement Specification (SRS), Problem-Solving, Project Management, Analytical Thinking, Testing & Validation, Business Relationship Management.

PROFESSIONAL EXPERIENCE

METLIFE <i>Business Data Analyst</i>	United States <i>August 2023 – Present</i>
• Utilized Python (Pandas, NumPy) and SQL (MySQL) to analyze large datasets, identified trends leading to 10% improvement in policy sales through targeted marketing. • Automated manual processes using Python scripts and SQL queries, reducing processing time by 30%. • Created interactive dashboards using Tableau to present complex data insights, enhancing decision-making by 25%. • Collaborated with the Compliance team to develop and implement data-driven monitoring tools for regulatory adherence. • Managed multiple IT projects using Agile and Jira, ensuring on-time delivery and a 20% increase in team productivity. • Developed and maintained comprehensive technical documentation to facilitate smooth knowledge transfer and reduce onboarding time by 40%.	

INTEL CORPORATION <i>System Validation Engineer Graduate Intern</i>	United States <i>June 2022 – May 2023</i>
• Developed and implemented an automation framework using Python and shell scripting to streamline pre-silicon testing workflows, utilizing regular expressions for real-time data extraction and analysis, and integrating an email module for automated reporting. • Developed and executed functional validation test cases for Intel CPU cores in C++, targeting power management features like C-states, P-states, and clock gating. • Debugged critical failures and optimized code to ensure test success, contributing to a 5% improvement in overall system stability (based on internal team metrics). • Automated backend workflows with Git, enabling efficient build and deployment pipelines and improving task completion times by 25%. • Performed BIOS Boot and Reset simulations on Intel FPGA platforms to validate UEFI booting across multiple operating systems. • Designed and validated performance test cases for CPU cores using C++, including workload simulations to assess throughput and power efficiency, resolving issues that improved performance by 10%. • Utilized JTAG, hardware boards, and lab equipment like oscilloscopes for debugging hardware-software integration, identifying and fixing system-level issues during validation.	

- Employed Use Case diagrams and User Story development to capture requirements for a new digital platform, ensuring 95% stakeholder satisfaction with the final product.
- Utilized Use Case diagrams and Flowcharts to map and analyze existing business processes, identifying 20% reduction potential in operational costs.
- Analyzed sales and customer data using MySQL and Excel, providing insights that informed 10% increase in targeted marketing campaigns.
- Developed and maintained dashboards for key performance indicators (KPIs), enhancing real-time decision-making capabilities.
- Coordinated with internal stakeholders (Operations, IT, and Sales) to ensure seamless project execution, resulting in zero project delays.
- Identified and documented operational risks using Risk Matrix and Heat Maps; collaborated with the Risk team to integrate controls to mitigate 80% of high-priority risks into the overall risk management framework.

PROJECTS

Linux Task Manager and Keyboard Logger

2021

- Developed a Linux Task Manager which display the Linux kernel information with live updates.
- The information was collected from /proc filesystem in Linux kernel and program was written in Python.
- The keyboard logger was able to display the information of different keystrokes.

Skills: Proc Filesystem, Python, Linux

Time Series Forecasting

2021

- Worked with time series data to predict stock price.
- Models used: fully connected neural network, long short-term memory, CNN.
- LSTM and CNN gave best RMSE score of 2.01 and 0.85 and worked as prominent technique for time series data.

Skills: Python, CNN, Pandas, Scikit-learn, NumPy

Big data handling

2020

- Reduced the size of the data using Hadoop's MapReduce and handled data manipulation tasks in HBase.
- Learnt HBase queries and filters to manipulate and simplify the data in larger size.

Skills: Hadoop Framework, HBase